

Projects

Astronomy Research

Lawrence, Kansas

Advisor: Dr. Elisabeth Mills

January 2023-April 2025

- Utilized Python and several packages, such as NumPy, Matplotlib, SciPy and Pandas to identify chemicals in a nebula through Gaussian fitting.
- Presented weekly updates on progress to collaborate with team members.

Class Research Project

The University of Kansas, Astr 591, Stellar Astronomy

- Fit model to data for several star clusters to determine each cluster's age in Python.

Observational Astrophysics Observing Project

The University of Kansas, Astr 596, Observational Astrophysics

- Remotely gathered data on star clusters and determined age, distance, and chemical makeup of observations in python.

Determining Muon Lifetime

The University of Kansas, Physics 616, Physical Measurements

- Muon lifetime was determined through fitting current models for muon decay to observed data in **Python** with **SciPy** and **Pandas**

Education

Bachelors of Science: Astronomy

Lawrence, Kansas

The University of Kansas

Fall 2021-May 2026

Honors College

Bachelors of Science: Physics

Lawrence, Kansas

The University of Kansas

Fall 2021-July 2026

Skills

Programming Languages Python, C++, HTML, CSS

Python Packages:SciPy, Numpy, Pandas, Matplotlib

Computer Skills:Git, Linux, TMUX

LanguagesEnglish (Fluent), Spanish (Conversational)

Professional Experience

ALMA Large Program ACES Team Meeting

Cambridge, Massachusetts

Harvard University

August 2024

- Presented research and worked with collaborators to work with data collected from the ALMA Large Program ACES (ALMA Central Molecular Zone Exploration Survey).

Python Package Astrodendro Contribution

November 2024

- Recognized bug in the astrodendro package and opened a pull request that was merged.